

**ISC 2026 EXAMINATION**  
**Sample Question Paper - 4**  
**Mathematics**

**Time Allowed: 3 hours**

**Maximum Marks: 80**

**General Instructions:**

*This Question Paper consists of three sections A, B and C.*

*Candidates are required to attempt all questions from **Section A** and all questions*

***EITHER** from **Section B OR Section C**.*

***Section A:** Internal choice has been provided in **two questions of two marks each, two questions of four marks each and two questions of six marks each**.*

***Section B:** Internal choice has been provided in **one question of two marks and one question of four marks**.*

***Section C:** Internal choice has been provided in **one question of two marks and one question of four marks**.*

*All working, including rough work, should be done on the same sheet as, and adjacent to the rest of the answer.*

*The intended marks for questions or parts of questions are given in brackets [ ].*

***Mathematical tables and graph papers are provided.***

**SECTION A - 65 MARKS**

1. In subparts (i) to (x) choose the correct options and in subparts (xi) to (xv), answer the questions as instructed. [15]

- (a) A square matrix A can be expressed as  $A = \frac{1}{2}(A + A') + \frac{1}{2}(A - A')$ , where [1]

- |                                                          |                                                          |
|----------------------------------------------------------|----------------------------------------------------------|
| a) $\frac{1}{2}(A + A')$ is a skew-symmetric matrix of A | b) $\frac{1}{2}(A - A')$ is a skew-symmetric matrix of A |
| c) $\frac{1}{2}(A' - A)$ is a symmetric matrix of A      | d) $\frac{1}{2}(A - A')$ is a symmetric matrix of A      |

- (b)  $\int_0^{2\pi} |\sin x| dx = ?$  [1]

- |      |      |
|------|------|
| a) 3 | b) 2 |
| c) 1 | d) 4 |

- (c) If  $\cot^{-1} \frac{1}{5} + \tan^{-1} x = \frac{\pi}{2}$ , then value of x is: [1]

- |      |                   |
|------|-------------------|
| a) 0 | b) $-\frac{1}{5}$ |
| c) 1 | d) $\frac{1}{5}$  |

- (d) The degree and order respectively of the differential equation  $\frac{dy}{dx} = \frac{1}{x+y+1}$  are [1]

- |         |         |
|---------|---------|
| a) 2, 1 | b) 1, 1 |
| c) 2, 2 | d) 1, 2 |

- (e) If A and B are two independent events such that  $P(A) = 0.3$ ,  $P(A \cup B) = 0.5$ , then  $P(A / B) - P(B / A) =$  [1]

a)  $\frac{2}{7}$

b)  $\frac{1}{7}$

c)  $\frac{3}{35}$

d)  $\frac{1}{70}$

- (f) Let R be a relation on set N given by  $R = \{(a, b) : a = b - 2, b > 6\}$ . Then [1]

a)  $(6, 8) \in R$

b)  $(2, 4) \in R$

c)  $(3, 8) \in R$

d)  $(8, 7) \in R$

- (g) The derivative of an even function is [1]

a) an odd function

b) negative

c) an even function

d) Positive

- (h) If  $e^{x+y} = xy$  then  $\frac{dy}{dx} = ?$  [1]

a)  $\frac{(x-xy)}{(xy-y)}$

b)  $\frac{1}{2} \frac{(x-xy)}{(xy-y)}$

c)  $\frac{y(1-x)}{x(y-1)}$

d)  $\frac{x(1-y)}{y(x-1)}$

- (i) Which of the following is not correct? [1]

a) If A is a skew-symmetric matrix of odd order, then  $|A| = 0$

b)  $|kA| = k^3 |A|$ , where  $A = [a_{ij}]_{3 \times 3}$

c)  $|A| = |A^T|$ , where  $A = [a_{ij}]_{3 \times 3}$

d)  $\begin{vmatrix} a+b & c+d \\ e+f & g+h \end{vmatrix} = \begin{vmatrix} a & c \\ e & g \end{vmatrix} + \begin{vmatrix} b & d \\ f & h \end{vmatrix}$

- (j) Let A and B be two symmetric matrices of order 3. [1]

**Assertion (A):**  $A(BA)$  and  $(AB)A$  are symmetric matrices.

**Reason (R):** AB is symmetric matrix, if matrix multiplication of A with B is commutative.

a) Both A and R are true and R is the correct explanation of A.

b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

- (k) Let  $S = \{1, 2, 3\}$  Determine whether the function  $f: S \rightarrow S$  defined as below have inverse. [1]  
Find  $f^{-1}$ , if it exists.

$$f = \{(1, 2) (2, 1) (3, 1)\}$$

- (l) If  $3A - B = \begin{bmatrix} 5 & 0 \\ 1 & 1 \end{bmatrix}$  and  $B = \begin{bmatrix} 4 & 3 \\ 2 & 5 \end{bmatrix}$ , then find the matrix A. [1]

- (m) Let  $f: R \rightarrow R$  be the function defined by  $f(x) = 4x - 3 \forall x \in R$ . Then write  $f^{-1}$  [1]

- (n) A coin is tossed three times. Let the events A and B be defined as follows: A = first toss is head, B = second toss is head. Check the independence of A and B. [1]

- (o) If  $E_1$  and  $E_2$  are independent events such that  $P(E_1) = 0.3$  and  $P(E_2) = 0.4$ , find [1]

$$P(\bar{E}_1 \cap E_2).$$

2. Differentiate  $\tan^{-1}\left(\frac{3x-x^3}{1-x^2}\right)$ , if  $x > \frac{1}{\sqrt{3}}$  [2]

OR

A stone is dropped into a quiet lake and waves move in circles at the speed of 5 cm/s. At the instant

when the radius of the circular wave is 8 cm, how fast is the enclosed area increasing?

3. Evaluate:  $\int \frac{dx}{(\sin x - \sin 2x)}$ . [2]
4. Find the intervals in which  $f(x) = \frac{x}{\log x}$  is increasing or decreasing. [2]
5. Evaluate the Integral:  $\int \frac{1}{x\sqrt{x^4-1}} dx$  [2]

OR

Evaluate:  $\int \frac{\tan x \sec^2 x}{(1 - \tan^2 x)} dx$ .

6. Let  $R_+$  be the set of all positive real numbers. Show that the function [2]
- $f: R_+ \rightarrow [-5, \infty) : f(x) = 9x^2 + 6x - 5$  invertible Find  $f^{-1}$
7. Find the value of  $\sin\left(2\tan^{-1}\frac{2}{3}\right) + \cos\left(\tan^{-1}\sqrt{3}\right)$  [4]
8. Evaluate the integral:  $\int \frac{x^2+x+1}{(x+1)^2(x+2)} dx$  [4]
9. Let  $f(x+y) = f(x) + f(y)$  for all  $x, y \in R$ . If  $f(x)$  is continuous at  $x = 0$ , show that  $f(x)$  is continuous at all  $x$ . [4]

OR

If  $x = a \cdot \sin(2t) \cdot (1 + \cos 2t)$  and  $y = b \cdot \cos 2t \cdot (1 - \cos 2t)$ , show that  $\left(\frac{dy}{dx}\right)_{at t=\frac{\pi}{4}} = \frac{b}{a}$

10. **Read the text carefully and answer the questions:** [4]

Mr. Ajay is taking up subjects of mathematics, physics, and chemistry in the examination. His probabilities of getting a grade A in these subjects are 0.2, 0.3, and 0.5 respectively.



- (a) Find the probability that Ajay gets Grade A in all subjects.
- (b) Find the probability that he gets Grade A in no subjects.
- (c) Find the probability that he gets Grade A in two subjects.
- (d) Find the probability that he gets Grade A in at least one subject.

OR

**Read the text carefully and answer the questions:**

[4]

Mr. Ajay is taking up subjects of mathematics, physics, and chemistry in the examination. His probabilities of getting a grade A in these subjects are 0.2, 0.3, and 0.5 respectively.



- (a) Find the value of  $x$ .

- (b) Find the value of  $y$ .
- (c) Find  $P\left(\frac{C}{B}\right)$ .
- (d) Find the probability that a randomly selected person of the society does Yoga of type A or B but not C.

11. **Read the text carefully and answer the questions:**

[6]

Three shopkeepers A, B and C go to a store to buy stationery. A purchase 12 dozen notebooks, 5 dozen pens and 6 dozen pencils. B purchases 10 dozen notebooks, 6 dozen pens and 7 dozen pencils. C purchases 11 dozen notebooks, 13 dozen pens and 8 dozen pencils. A notebook costs ₹40, a pen costs ₹12 and a pencil costs ₹3.



- (a) How are the number of items purchased by shopkeepers A, B, and C represented in matrix form?
- (b) If  $X$  represents a matrix, and  $Y$  represents the matrix formed by the cost of each item, what does the product  $XY$  equal?
- (c) If  $A^2 = A$ , then what is the value of  $(A + I)^3 - 7A$ ?

12. Solve  $(x^2 - y^2)dx + 2xydy = 0$

[6]

OR

Show that the differential equation of  $(x^2 + xy)dy = (x^2 + y^2)dx$  is homogeneous and solve it.

13. Show that the curves intersect orthogonally at the indicated points  $y^2 = 8x$  and  $2x^2 + y^2 = 10$  at  $(1, 2\sqrt{2})$ .

[6]

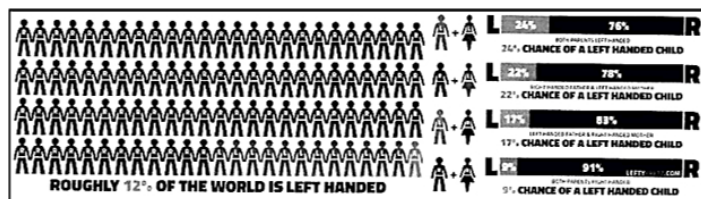
OR

Show that the right circular cone of least curved surface and given volume has an altitude equal to  $\sqrt{2}$  times the radius of the base.

14. **Read the text carefully and answer the questions:**

[6]

Recent studies suggest that roughly 12% of the world population is left handed.



Depending upon the parents, the chances of having a left handed child are as follows:

- A. When both father and mother are left handed:  
Chances of left handed child is 24%.
- B. When father is right handed and mother is left handed:  
Chances of left handed child is 22%.
- C. When father is left handed and mother is right handed:  
Chances of left handed child is 17%.
- D. When both father and mother are right handed:  
Chances of left handed child is 9%.

Assuming that  $P(A) = P(B) = P(C) = P(D) = \frac{1}{4}$  and  $L$  denotes the event that child is left handed.

- (a) Find  $P\left(\frac{L}{C}\right)$
- (b) Find  $P\left(\frac{L}{A}\right)$
- (c) Find  $P\left(\frac{A}{L}\right)$
- (d) Find the probability that a randomly selected child is left handed given that exactly one of the parents is left handed.

#### SECTION B - 15 MARKS

15. In subparts (i) and (ii) choose the correct options and in subparts (iii) to (v), answer the questions as instructed. [5]

- (a) If  $\vec{a} \cdot \vec{b} = \frac{1}{2} |\vec{a}| |\vec{b}|$ , then the angle between  $\vec{a}$  and  $\vec{b}$  is [1]

- a)  $90^\circ$
- b)  $30^\circ$
- c)  $0^\circ$
- d)  $60^\circ$

- (b) Find the angle between the pair of lines [1]

$$\frac{x+3}{3} = \frac{y-1}{5} = \frac{z+3}{4}$$

$$\frac{x+1}{1} = \frac{y-4}{1} = \frac{z-5}{2}$$

- (c) Find a vector of magnitude 9, which is perpendicular to both the vectors  $4\hat{i} - \hat{j} + 3\hat{k}$  and  $-2\hat{i} + \hat{j} - 2\hat{k}$ . [1]

- (d) If a line makes an angle of  $30^\circ$  with the positive direction of x-axis,  $120^\circ$  with the positive direction of y-axis, then the angle which it makes with the positive direction of z-axis is: [1]

- a)  $0^\circ$
- b)  $120^\circ$
- c)  $90^\circ$
- d)  $60^\circ$

- (e) Find the value of  $p$ , so that lines  $\frac{x-1}{-2} = \frac{y-4}{3p} = \frac{z-3}{4}$  and  $\frac{x-2}{4p} = \frac{y-5}{2} = \frac{1-z}{7}$  are perpendicular to each other. [1]

16. If  $\vec{a} = \hat{i} + 2\hat{j} - 3\hat{k}$  and  $\vec{b} = 3\hat{i} - \hat{j} + 2\hat{k}$ , show that  $(\vec{a} + \vec{b})$  is perpendicular  $(\vec{a} - \vec{b})$ . [2]

OR

Find the angle between the vectors  $\vec{a}$  and  $\vec{b}$ , where  $\vec{a} = \hat{i} - \hat{j}$  and  $\vec{b} = \hat{j} + \hat{k}$

17. Find the angle between the planes whose vector equations are  $\vec{r} \cdot (2\hat{i} + 2\hat{j} - 3\hat{k}) = 5$  and  $\vec{r} \cdot (3\hat{i} - 3\hat{j} + 5\hat{k}) = 3$ . [4]

OR

Find the image of the point  $(3, -2, 1)$  in the plane  $3x - y + 4z = 2$ .

18. Make a sketch of the region  $\{(x, y) : 0 \leq y \leq x^2 + 3; 0 \leq y \leq 2x + 3; 0 \leq x \leq 3\}$  and find its area using integration. [4]

#### SECTION C - 15 MARKS

19. In subparts (i) and (ii) choose the correct options and in subparts (iii) to (v), answer the questions as instructed. [5]

- (a) If the cost function of a certain commodity is  $C(x) = 2000 + 50x - \frac{1}{5}x^2$ , then the average [1]

cost of producing 5 units is

- a) ₹ 2245
- b) ₹ 451
- c) ₹ 449
- d) ₹ 450

(b) A linear programming problem deals with the optimization of a/an: [1]

- a) linear function
- b) logarithmic function
- c) quadratic function
- d) exponential function

(c) Two regression lines are represented by  $2x + 3y - 10 = 0$  and  $4x + y - 5 = 0$ . Find the line of regression of  $y$  on  $x$ . [1]

(d) A company has fixed costs of ₹ 26,000. The cost of producing one item is ₹ 30. If this item sells for ₹ 43, find the break-even point. [1]

(e) A firm knows that the demand function for one of its products is linear. It also knows that it can sell 1400 units when the price is ₹ 4 per unit and it can sell 1800 units at a price ₹ 2 per unit. Find the marginal revenue function of this product. [1]

20. A firm knows that the demand function for one of its products is linear. It also knows that it can sell 1000 units when the price is ₹ 4 per unit, and it can sell 1500 units when the price is ₹ 2a unit. Determine: [2]

- i. the demand function
- ii. the total revenue function
- iii. the average revenue function
- iv. the marginal revenue function

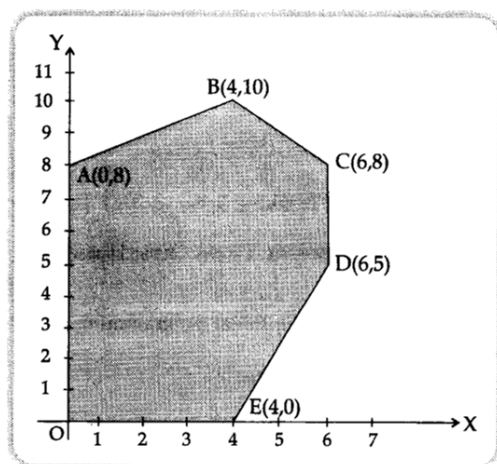
OR

Suppose the cost to produce some commodity is a linear function of output. Find cost as a function of output, if costs are ₹ 4000 for 250 units and ₹ 5000 for 350 units.

21. By using the data  $\bar{x} = 25$ ,  $\bar{y} = 30$ ,  $b_{yx} = 1.6$  and  $b_{xy} = 0.4$ , find: [4]

- i. The regression equation  $y$  on  $x$ .
- ii. What is the most likely value of  $y$  when  $x = 60$ ?
- iii. What is the coefficient of correlation between  $x$  and  $y$ ?

22. The corner points of the feasible region determined by the system of linear constraints are as shown below: [4]



Answer each of the following:

- i. Let  $Z = 3x - 4y$  be the objective function. Find the maximum and minimum value of  $Z$  and also the corresponding points at which the maximum and minimum value occurs.
- ii. Let  $Z = px + qy$ , where  $p, q > 0$  be the objective function. Find the condition on  $p$  and  $q$  so that the maximum value of  $Z$  occurs at  $B(4, 10)$  and  $C(6, 8)$ . Also mention the number of optimal solutions in this case.

OR

A gardener has a supply of fertilizers of the type I which consists of 10% nitrogen and 6% phosphoric acid, and of the type II which consists of 5% nitrogen and 10% phosphoric acid. After testing the soil conditions, he finds that he needs at least 14 kg of nitrogen and 14 kg of phosphoric acid for his crop. If the type-I fertilizer costs 60 paise per kg and the type-II fertilizer costs 40 paise per kg, determine how many kilograms of each type of fertilizer should be used so that the nutrient requirements are met at a minimum cost. What is the minimum cost?

# Solution

## SECTION A - 65 MARKS

1. In subparts (i) to (x) choose the correct options and in subparts (xi) to (xv), answer the questions as instructed.

- (a) **(b)**  $\frac{1}{2}(A - A')$  is a skew-symmetric matrix of A

**Explanation:**

$$\text{Let } B = \frac{1}{2}(A - A')$$

$$B' = \frac{1}{2}(A - A')' = \frac{1}{2}[A' - (A')']$$

$$= \frac{1}{2}(A' - A) = -\frac{1}{2}(A - A') = -B$$

$$\therefore \frac{1}{2}(A - A') \text{ is skew-symmetric matrix of A.}$$

- (b) **(d)** 4

**Explanation:**

$$\text{Let the given integral be } = \int_0^{2\pi} |\sin x| dx$$

Now, let us find the equivalent expression to  $|\sin x|$  at  $0 \leq x \leq 2\pi$

$$\text{In } 0 \leq x \leq \pi$$

$$|\sin x| = \sin x$$

$$\text{In } \pi \leq x \leq 2\pi$$

$$|\sin x| = -\sin x$$

$$\therefore I = \int_0^{\pi} \sin x dx + \int_{\pi}^{2\pi} -\sin x dx$$

$$= -\cos \pi - (-\cos 0) + \cos 2\pi - \cos \pi$$

$$= -(-1) + 1 + 1 - (-1)$$

$$= 2 + 2$$

$$= 4$$

- (c) **(d)**  $\frac{1}{5}$

**Explanation:**

$$\cot^{-1} \frac{1}{5} + \tan^{-1} x = \frac{\pi}{2}$$

$$\text{Also, } \cot^{-1} x + \tan^{-1} x = \frac{\pi}{2}$$

$$\text{From (i) \& (ii), we get } x = \frac{1}{5}$$

- (d) **(b)** 1, 1

**Explanation:**

$$1, 1$$

- (e) **(d)**  $\frac{1}{70}$

**Explanation:**

$$P(A) = 0.3, P(A \cup B) = 0.5 \text{ (Given)}$$

Since, A and B are two independent events,

$$P(A \cap B) = P(A) P(B)$$

$$P(A \cap B) = 0.3 \times P(B) \dots (i)$$

Also, according to the addition theorem of probability,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$0.5 = 0.3 + P(B) - 0.3 P(B) \text{ From (Given) \& (i)}$$

$$0.7 P(B) = 0.2$$

$$P(B) = \frac{0.2}{0.7} = \frac{2}{7} \dots (ii)$$

Putting value of P(B) in equation (i) we get,

$$P(A \cap B) = 0.3 \times \frac{2}{7} = \frac{3}{10} \times \frac{2}{7}$$

$$P(A \cap B) = \frac{6}{70} \dots (iii)$$

Now,

$$\begin{aligned} P\left(\frac{A}{B}\right) - P\left(\frac{B}{A}\right) &= \frac{P(A \cap B)}{P(B)} - \frac{P(A \cap B)}{P(A)} \\ &= \frac{\frac{6}{70}}{\frac{2}{3}} - \frac{\frac{6}{70}}{\frac{10}{3}} \text{ From (iii) \& (ii) and (Given)} \\ &= \frac{6}{70} \times \frac{7}{2} - \frac{6}{70} \times \frac{10}{3} \\ &= \frac{3}{10} - \frac{2}{7} \\ &= \frac{1}{70} \end{aligned}$$

- (f) **(a)**  $(6, 8) \in R$

**Explanation:**

For option  $(6, 8)$

$$b = 8 > 6$$

$$\text{also, } b = 8 - 2 = 6 \Rightarrow a = b - 2$$

- (g) **(a)** an odd function

**Explanation:**

The derivative of an even function is an odd function.

- (h) **(c)**  $\frac{y(1-x)}{x(y-1)}$

**Explanation:**

$$\text{Given that } xy = e^{x+y}$$

Taking log both sides, we get

$$\log_e xy = x + y \text{ (Since } \log_a b^c = c \log_a b \text{)}$$

Since  $\log_a bc = \log_a b + \log_a c$ , we get

$$\log_e x + \log_e y = x + y$$

Differentiating with respect to  $x$ , we obtain

$$\frac{1}{x} + \frac{1}{y} \frac{dy}{dx} = 1 + \frac{dy}{dx}$$

Or

$$\frac{dy}{dx} \left( \frac{y-1}{y} \right) = \frac{1-x}{x}$$

$$\text{Therefore, } \frac{dy}{dx} = \frac{y(1-x)}{x(y-1)}$$

- (i) **(d)**  $\begin{vmatrix} a+b & c+d \\ e+f & g+h \end{vmatrix} = \begin{vmatrix} a & c \\ e & g \end{vmatrix} + \begin{vmatrix} b & d \\ f & h \end{vmatrix}$

**Explanation:**

For adding the determinants, we need to find the value of the determinants and add them. We cannot apply the method applicable for matrix addition.

- (j) **(b)** Both A and R are true but R is not the correct explanation of A.

**Explanation:**

**Assertion:** Since, A and B are symmetric matrices.

$$\therefore A^T = A \text{ and } B^T = B.$$

Now, to check A (BA) is symmetric.

$$\begin{aligned} \text{Consider } [A(BA)]^T &= (BA)^T \cdot A^T = (A^T B^T) A^T \\ &= (AB)A = A(BA) \end{aligned}$$

$$\text{So, } [A(BA)]^T = A(BA)$$

$\Rightarrow A(BA)$  is symmetric.

Similarly, (AB) A is symmetric.

So, Assertion is true.

**Reason:** Now,  $(AB)' = B'A'$

$$= BA$$

This will be symmetric, if A and B is commutative i.e.  $AB = BA$ .

Hence, both Assertion and Reason are true but Reason is not the correct explanation of Assertion.

(k)  $f(2) = 1, f(3) = 1$ , as two elements of domain have same image in co-domain.

$f$  is not one - one so that  $f$  is not invertible

Hence no inverse.

$$(l) \text{ Given } 3A - B = \begin{bmatrix} 5 & 0 \\ 1 & 1 \end{bmatrix}, B = \begin{bmatrix} 4 & 3 \\ 2 & 5 \end{bmatrix}$$

$$\text{or } 3A - \begin{bmatrix} 4 & 3 \\ 2 & 5 \end{bmatrix} = \begin{bmatrix} 5 & 0 \\ 1 & 1 \end{bmatrix}$$

$$\text{or } 3A = \begin{bmatrix} 5 & 0 \\ 1 & 1 \end{bmatrix} + \begin{bmatrix} 4 & 3 \\ 2 & 5 \end{bmatrix}$$

$$\text{or } 3A = \begin{bmatrix} 9 & 3 \\ 3 & 6 \end{bmatrix}$$

$$\text{or } A = \begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix}$$

(m) Given that  $f(x) = 4x - 3 = y$  (say), then  $4x = y + 3$

$$\Rightarrow x = \frac{y+3}{4}$$

$$\text{Hence } f^{-1}(y) = \frac{y+3}{4} \Rightarrow f^{-1}(x) = \frac{x+3}{4}$$

(n) The sample space is given by,

$$S = \{(H H H), (H H T), (H T H), (H T T), (T H H), (T H T), (T T H), (T T T)\}$$

$$P(A) = \frac{4}{8} = \frac{1}{2}$$

$$P(B) = \frac{4}{8} = \frac{1}{2}$$

$$P(A \cap B) = \frac{2}{8} = \frac{1}{4} = P(A)P(B)$$

Thus, A and B are independent events.

(o) We are given that,  $E_1$  and  $E_2$  are two independent events such that  $P(E_1) = 0.3$  and  $P(E_2) = 0.4$

$$P(\overline{E_1} \cap E_2) = P(\overline{E_1}) \times P(E_2)$$

$$= 0.7 \times 0.4 = 0.28$$

$$\text{Thus, } P(\overline{E_1} \cap E_2) = 0.28$$

$$2. \text{ Let } y = \tan^{-1} \left( \frac{3x-x^3}{1-3x^2} \right).$$

Put,  $x = \tan \theta$ , we get

$$y = \tan^{-1} \left( \frac{3 \tan \theta - \tan^3 \theta}{1 - 3 \tan^2 \theta} \right) = \tan^{-1}(\tan 3\theta)$$

If  $x > \frac{1}{\sqrt{3}}$ , then

$$x = \tan \theta \Rightarrow \tan \theta > \frac{1}{\sqrt{3}} \Rightarrow \frac{\pi}{6} < \theta < \frac{\pi}{2} \Rightarrow \frac{\pi}{2} < 3\theta < \frac{3\pi}{2}$$

$$\therefore y = \tan^{-1}(\tan 3\theta)$$

$$\Rightarrow y = \tan^{-1}(-\tan(\pi - 3\theta))$$

$$\Rightarrow y = \tan^{-1}\{\tan(3\theta - \pi)\}$$

$$\Rightarrow y = 3\theta - \pi \quad \left[ \because \frac{\pi}{2} < 3\theta < \frac{3\pi}{2} \Rightarrow -\frac{\pi}{2} < 3\theta - \pi < \frac{\pi}{2} \right]$$

$$\Rightarrow y = 3 \tan^{-1} x - \pi \quad \left[ \because x = \tan \theta \Rightarrow \theta = \tan^{-1} x \right]$$

$$\Rightarrow \frac{dy}{dx} = \frac{3}{1+x^2} - 0 = \frac{3}{1+x^2}$$

OR

Let  $x$  cm be the radius and  $y$  be the enclosed area of the circular wave at any time  $t$ .

Rate of increase of radius of circular wave = 5 cm/sec

$$\Rightarrow \frac{dx}{dt} \text{ is positive and } = 5 \text{ cm/sec}$$

$$\Rightarrow \frac{dx}{dt} = 5 \text{ cm/sec} \dots (i)$$

$$y = \pi x^2$$

$$\therefore \text{Rate of change of area} = \frac{dy}{dt} = \pi \frac{d}{dt} x^2$$

$$= \pi \cdot 2x \frac{dx}{dt} = 2\pi x (5) \text{ (from (i))}$$

$$= 10\pi x \text{ cm}^2 / \text{sec}$$

Putting  $x = 8 \text{ cm}$  (given),

$$\frac{dy}{dt} = 10\pi (8) = 80\pi \text{ cm}^2 / \text{sec}$$

Since  $\frac{dy}{dt}$  is positive, therefore area of circular wave is increasing at the rate of  $80\pi \text{ cm}^2 / \text{sec}$ .

3. Let  $I = \int \frac{dx}{(\sin x - \sin 2x)}$ , then

$$I = \int \frac{dx}{(\sin x - 2 \sin x \cos x)}$$

$$= \int \frac{dx}{\sin x (1 - 2 \cos x)} = \int \frac{\sin x}{\sin^2 x (1 - 2 \cos x)} dx$$

$$= - \int \frac{dt}{(1-t^2)(1-2t)}, \text{ where } \cos x = t$$

$$= \int \frac{dt}{(t-1)(t+1)(1-2t)} \dots\dots (i)$$

$$\text{Let } \frac{1}{(t-1)(t+1)(1-2t)} = \frac{A}{(t-1)} + \frac{B}{(t+1)} + \frac{C}{(1-2t)}$$

$$\text{Then, } 1 = A(t+1)(1-2t) + B(t-1)(1-2t) + C(t-1)(t+1) \dots\dots (ii)$$

$$\text{Putting } t = 1 \text{ in (ii), we get } A = \frac{-1}{2}$$

$$\text{Putting } t = -1 \text{ in (ii), we get } B = \frac{-1}{6}$$

$$\text{Putting } t = \frac{1}{2} \text{ in (ii), we get } C = \frac{-4}{3}$$

$$\therefore I = -\frac{1}{2} \int \frac{dt}{(t-1)} - \frac{1}{6} \int \frac{dt}{(t+1)} - \frac{4}{3} \int \frac{dt}{(1-2t)}$$

$$= -\frac{1}{2} \log |t-1| - \frac{1}{6} \log |t+1| + \frac{2}{3} \int \frac{-2dt}{(1-2t)}$$

$$= -\frac{1}{2} \log |t-1| - \frac{1}{6} \log |t+1| + \frac{2}{3} \log |1-2t| + C$$

$$= -\frac{1}{2} \log |\cos x - 1| - \frac{1}{6} \log |\cos x + 1| + \frac{2}{3} \log |1 - 2 \cos x| + C$$

4. Note that the domain of  $f(x)$  is the set of all positive real numbers other than unity ie;  $(0,1) \cup (1, \infty)$

$$\text{Now } f(x) = \frac{x}{\log x}$$

$$\Rightarrow f'(x) = \frac{\log x - 1}{(\log x)^2}$$

For  $f(x)$  to be increasing function, we must have

$$f'(x) > 0$$

$$\Rightarrow \frac{\log x - 1}{(\log x)^2} > 0$$

$$\Rightarrow \log x - 1 > 0$$

$$\log x > 1$$

$$x > e^1$$

So,  $f(x)$  is increasing on  $(e, \infty)$

For  $f(x)$  to be decreasing we must have

$$\Rightarrow \frac{\log x - 1}{(\log x)^2} < 0$$

$$\Rightarrow \log x - 1 < 0$$

$$\Rightarrow \log x < 1$$

$$\Rightarrow x < e^1$$

So  $f(x)$  is decreasing on  $(0, e)$

5. Let  $I = \int \frac{1}{x\sqrt{x^4-1}} dx$

$$\text{Since } \int \frac{1}{x\sqrt{x^2-1}} dx = \sec^{-1} x + c$$

$$\text{We have, } I = \int \frac{1}{x\sqrt{x^4-1}} dx \dots\dots\dots (i)$$

Multiplying numerator and denominator with  $x$

$$I = \int \frac{x}{x^2 \sqrt{(x^2)^2 - 1}} dx$$

Let  $x^2 = t$

$$\Rightarrow 2x = \frac{dt}{dx}$$

$$\Rightarrow x dx = \frac{dt}{2}$$

Putting this value in equation (i), we get

$$I = \frac{1}{2} \int \frac{dt}{t \sqrt{t^2 - 1}} [x^2 = t]$$

$$\Rightarrow I = \frac{1}{2} \sec^{-1} t + c$$

$$\Rightarrow I = \frac{1}{2} \sec^{-1} (x^2) + c$$

OR

Take  $\tan x = a$

Hence,  $\sec^2 x dx = da$

$$\therefore \int \frac{\tan x \sec^2 x}{1 - \tan^2 x} dx$$

$$= \int \frac{ada}{1 - a^2}$$

Now, taking  $1 - a^2 = k$ ,  $-2a da = dk$  i.e.  $a da = -dk/2$

$$\therefore \int \frac{ada}{1 - a^2}$$

$$= \int \frac{-dk}{2k}$$

$$= -\frac{1}{2} \ln |k| + c$$

Replacing the value of  $k$ ,

$$-\frac{1}{2} \ln |k| + c$$

$$= -\frac{1}{2} \ln |1 - a^2| + c$$

Replacing the value of  $a$ ,

$$-\frac{1}{2} \ln |1 - a^2| + c$$

$$= -\frac{1}{2} \ln |1 - \tan^2 x| + c$$

Where  $c$  is the integrating constant

6. Given,  $f(x) = 9x^2 + 6x - 5$

For,  $a, b \in \mathbb{R}_+$

$$\Rightarrow f(a) = f(b)$$

$$\Rightarrow 9a^2 + 6a - 5 = 9b^2 + 6b - 5$$

$$\Rightarrow a = b$$

This,  $f$  is a one one function

Now, Let  $y = f(x) = (9x^2 + 6x - 5)$

$$\text{So } x = \frac{-1 + \sqrt{y+6}}{3} \text{ [Range of } f(x) = \text{Domain of } y]$$

So Domain of  $y = \text{Range of } f(x) = [-5, \infty]$

Hence, Range of  $f(x) = \text{co-domain of } f(x) = [-5, \infty]$

So,  $f(x)$  is onto function

As it is bijective function.

So it is invertible

$$\text{Inverse of } f(x) \text{ is } f^{-1}(x) = \frac{-1 + \sqrt{x+6}}{3}$$

7. Let  $\tan^{-1} \frac{2}{3} = x$  and  $\tan^{-1} \sqrt{3} = y$

$$\text{so that } \tan x = \frac{2}{3} \text{ and } \tan y = \sqrt{3}$$

Therefore,

$$\sin(2 \tan^{-1} \frac{2}{3}) + \cos(\tan^{-1} \sqrt{3})$$

$$= \sin(2x) + \cos y$$

$$\begin{aligned}
&= \frac{2 \tan x}{1 + \tan^2 x} + \frac{1}{\sqrt{1 + \tan^2 y}} \\
&= \frac{2 \cdot \frac{2}{3}}{1 + \frac{4}{9}} + \frac{1}{\sqrt{1 + (\sqrt{3})^2}} \\
&= \frac{12}{13} + \frac{1}{2} = \frac{37}{26}
\end{aligned}$$

8. We have,

$$I = \int \frac{x^2 + x + 1}{(x+1)^2(x+2)} dx$$

To solve this we use partial fraction.

$$\text{Let } \frac{x^2 + x + 1}{(x+1)^2(x+2)} = \frac{A}{x+1} + \frac{B}{(x+1)^2} + \frac{C}{x+2} \dots(i)$$

$$\Rightarrow x^2 + x + 1 = A(x+1)(x+2) + B(x+2) + C(x+1)^2 \dots(ii)$$

Putting  $x = -1$  in (ii), we get

$$B = 1$$

Putting  $x = -2$  in (ii), we get

$$C = 3$$

Putting  $x = 0$  in (ii), we get

$$1 = 2A + 2B + C$$

$$\Rightarrow 1 = 2A + 2 + 3$$

$$\Rightarrow -4 = 2A$$

$$\Rightarrow A = -2$$

Now, (i) becomes

$$\frac{x^2 + x + 1}{(x+1)^2(x+2)} = \frac{-2}{x+1} + \frac{1}{(x+1)^2} + \frac{3}{x+2}$$

Therefore, integral becomes

$$\begin{aligned}
I &= \int \left[ \frac{-2}{x+1} + \frac{1}{(x+1)^2} + \frac{3}{x+2} \right] dx \\
&= -2 \log |x+1| - \frac{1}{(x+1)} + 3 \log |x+2| + C
\end{aligned}$$

9. Since  $f(x)$  is continuous at  $x = 0$ .

Therefore,

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^+} f(x) = f(0)$$

$$\Rightarrow \lim_{h \rightarrow 0} f(0-h) = \lim_{h \rightarrow 0} f(0+h) = f(0)$$

$$\Rightarrow \lim_{h \rightarrow 0} f(0+(-h)) = \lim_{h \rightarrow 0} f(0+h) = f(0)$$

$$\Rightarrow \lim_{h \rightarrow 0} [f(0) + f(-h)] = \lim_{h \rightarrow 0} [f(0) + f(h)] = f(0) \dots\dots[\text{Using: } f(x+y) = f(x) + f(y)]$$

$$\Rightarrow f(0) + \lim_{h \rightarrow 0} f(-h) = f(0) + \lim_{h \rightarrow 0} f(h) = f(0)$$

$$\Rightarrow \lim_{h \rightarrow 0} f(-h) = \lim_{h \rightarrow 0} f(h) = 0 \dots(i)$$

Let 'a' be any real number.

Then,

$$\lim_{x \rightarrow a^-} f(x) = \lim_{h \rightarrow 0} f(a-h) = \lim_{h \rightarrow 0} f(a+(-h))$$

$$\Rightarrow \lim_{x \rightarrow a^-} f(x) = \lim_{h \rightarrow 0} [f(a) + f(-h)] \dots\dots[\because f(x+y) = f(x) + f(y)]$$

$$\Rightarrow \lim_{x \rightarrow a^-} f(x) = f(a) + \lim_{h \rightarrow 0} f(-h)$$

$$\Rightarrow \lim_{x \rightarrow a^-} f(x) = f(a) + 0 \text{ [Using (i)]}$$

$$\Rightarrow \lim_{x \rightarrow a^-} f(x) = f(a)$$

$$\text{and, } \lim_{x \rightarrow a^+} f(x) = \lim_{h \rightarrow 0} f(a+h)$$

$$\Rightarrow \lim_{x \rightarrow a^+} f(x) = \lim_{h \rightarrow 0} [f(a) + f(h)] \dots [\because f(x+y) = f(x) + f(y)]$$

$$\Rightarrow \lim_{x \rightarrow a^+} f(x) = f(a) + \lim_{h \rightarrow 0} f(h)$$

$$\Rightarrow \lim_{x \rightarrow a^+} f(x) = f(a) + 0 = f(a) \quad \dots [\text{Using (i)}]$$

Thus, we have

$$\lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x) = f(a)$$

$\therefore f(x)$  is continuous at  $x = a$

Since 'a' is an arbitrary real number.

So,  $f(x)$  is continuous at all  $x \in \mathbb{R}$

OR

Here,

$$x = a \cdot \sin(2t) \cdot (1 + \cos 2t) \quad \text{and} \quad y = b \cdot \cos(2t) \cdot (1 - \cos 2t)$$

$$\therefore \frac{dx}{dt} = a \left[ \sin 2t \cdot \frac{d}{dt}(1 + \cos 2t) + (1 + \cos 2t) \cdot \frac{d}{dt} \sin 2t \right]$$

$$= a \left[ \sin 2t \cdot (-\sin 2t) \cdot \frac{d}{dt} 2t + (1 + \cos 2t) \cdot \cos 2t \cdot \frac{d}{dt} 2t \right]$$

$$= -2a \cdot \sin^2(2t) + 2a \cdot \cos(2t)(1 + \cos 2t)$$

$$\Rightarrow \frac{dx}{dt} = -2a [\sin^2 2t - \cos 2t(1 + \cos 2t)] \quad \dots (i)$$

$$\text{and } \frac{dy}{dt} = b \left[ \cos 2t \cdot \frac{d}{dt}(1 - \cos 2t) + (1 - \cos 2t) \cdot \frac{d}{dt} \cos 2t \right]$$

$$= b \left[ \cos 2t \cdot (\sin 2t) \frac{d}{dt} 2t + (1 - \cos 2t)(-\sin 2t) \cdot \frac{d}{dt} 2t \right]$$

$$= b[2\sin 2t \cdot \cos 2t + 2(1 - \cos 2t)(-\sin 2t)]$$

$$= b[2\sin 2t \cdot \cos 2t + 2(1 - \cos 2t)(-\sin 2t)] \quad \dots (ii)$$

$$\therefore \frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{-2b[-\sin 2t \cos 2t + (1 - \cos 2t) \sin 2t]}{-2a[\sin^2 2t - \cos 2t(1 + \cos 2t)]}$$

$$= \frac{b}{a} \cdot \frac{(0+1)}{(1-0)} \quad \dots \left[ \because \sin \frac{\pi}{2} = 1 \text{ and } \cos \frac{\pi}{2} = 0 \right]$$

$$= \frac{b}{a}$$

Hence proved.

#### 10. Read the text carefully and answer the questions:

Mr. Ajay is taking up subjects of mathematics, physics, and chemistry in the examination. His probabilities of getting a grade A in these subjects are 0.2, 0.3, and 0.5 respectively.



- (a)  $P(\text{Grade A in Maths}) = P(M) = 0.2$   
 $P(\text{Grade A in Physics}) = P(P) = 0.3$   
 $P(\text{Grade A in Chemistry}) = P(C) = 0.5$   
 $P(\text{Grade A in all subjects}) = P(M \cap P \cap C) = P(M) \cdot P(P) \cdot P(C)$   
 $P(\text{Grade A in all subjects}) = 0.2 \times 0.3 \times 0.5 = 0.03$
- (b)  $P(\text{Grade A in Maths}) = P(M) = 0.2$   
 $P(\text{Grade A in Physics}) = P(P) = 0.3$   
 $P(\text{Grade A in Chemistry}) = P(C) = 0.5$   
 $P(\text{Grade A in no subjects}) = P(\bar{M} \cap \bar{P} \cap \bar{C}) = P(\bar{M}) \cdot P(\bar{P}) \cdot P(\bar{C})$   
 $P(\text{Grade A in no subjects}) = 0.8 \times 0.7 \times 0.5 = 0.280$
- (c)  $P(\text{Grade A in Maths}) = P(M) = 0.2$   
 $P(\text{Grade A in Physics}) = P(P) = 0.3$   
 $P(\text{Grade A in Chemistry}) = P(C) = 0.5$   
 $P(\text{Grade A in 2 subjects}) = P(M \cap P \cap \bar{C}) + P(P \cap C \cap \bar{M}) + P(M \cap C \cap \bar{P})$

$$P(\text{Grade A in 2 subjects}) = 0.2 \times 0.3 \times 0.5 + 0.3 \times 0.5 \times 0.8 + 0.2 \times 0.5 \times 0.7 = 0.03 + 0.12 + 0.07 = 0.22$$

$$P(\text{Grade A in 2 subjects}) = 0.22$$

$$(d) P(\text{Grade A in Maths}) = P(M) = 0.2$$

$$P(\text{Grade A in Physics}) = P(P) = 0.3$$

$$P(\text{Grade A in Chemistry}) = P(C) = 0.5$$

$$P(\text{Grade A in atleast one subject}) = 1 - P(\text{grade A in no subject}) = 1 - P(\bar{M} \cap \bar{P} \cap \bar{C})$$

$$P(\text{Grade A in atleast one subjects}) = 1 - 0.280 = 0.72$$

OR

**Read the text carefully and answer the questions:**

Mr. Ajay is taking up subjects of mathematics, physics, and chemistry in the examination. His probabilities of getting a grade A in these subjects are 0.2, 0.3, and 0.5 respectively.



$$(a) x + 0.21 = 0.44 \Rightarrow x = 0.23$$

$$(b) 0.41 + y + 0.44 + 0.11 = 1 \Rightarrow y = 0.04$$

$$(c) P\left(\frac{C}{B}\right) = \frac{P(C \cap B)}{P(B)}$$

$$P(B) = 0.09 + 0.04 + 0.23 = 0.36$$

$$P\left(\frac{C}{B}\right) = \frac{0.23}{0.36} = \frac{23}{36}$$

$$(d) P(A \text{ or } B \text{ but not } C)$$

$$= 0.32 + 0.09 + 0.04$$

$$= 0.45$$

#### 11. Read the text carefully and answer the questions:

Three shopkeepers A, B and C go to a store to buy stationery. A purchase 12 dozen notebooks, 5 dozen pens and 6 dozen pencils. B purchases 10 dozen notebooks, 6 dozen pens and 7 dozen pencils. C purchases 11 dozen notebooks, 13 dozen pens and 8 dozen pencils. A notebook costs ₹40, a pen costs ₹12 and a pencil costs ₹3.



(a) Number of items purchased by shopkeepers A, B and C can be written in matrix form as

$$X = \begin{matrix} & \begin{matrix} \text{Notebooks} & \text{pens} & \text{pencils} \end{matrix} \\ \begin{matrix} A \\ B \\ C \end{matrix} & \begin{bmatrix} 144 & 60 & 72 \\ 120 & 72 & 84 \\ 132 & 156 & 96 \end{bmatrix} \end{matrix}$$

$$(b) \text{ Since, } Y = \begin{bmatrix} 40 \\ 12 \\ 3 \end{bmatrix} \begin{matrix} \text{Note book} \\ \text{Pen} \\ \text{Pencil} \end{matrix}$$

$$\therefore XY = \begin{bmatrix} 144 & 60 & 72 \\ 120 & 72 & 84 \\ 132 & 156 & 96 \end{bmatrix} \begin{bmatrix} 40 \\ 12 \\ 3 \end{bmatrix}$$

$$= \begin{bmatrix} 5760 + 720 + 216 \\ 4800 + 864 + 252 \\ 5280 + 1872 + 288 \end{bmatrix} = \begin{bmatrix} 6696 \\ 5916 \\ 7440 \end{bmatrix}$$

$$(c) (A + I)^2 = A^2 + 2A + I = 3A + I$$

$$\Rightarrow (A + I)^3 = (3A + I)(A + I)$$

$$= 3A^2 + 4A + I = 7A + I$$

$$\therefore (A + I)^3 - 7A = I$$

$$12. (x^2 - y^2)dx + 2xy dy = 0$$

$$\Rightarrow (x^2 - y^2)dx = -2xy dy$$

$$\Rightarrow \frac{dy}{dx} = \frac{y^2 - x^2}{2xy}$$

$$\Rightarrow \frac{dy}{dx} = \frac{\left(\frac{y}{x}\right)^2 - 1}{2\left(\frac{y}{x}\right)} \quad (1)$$

$$\text{Put } y = vx, \text{ then, } \frac{dy}{dx} = v + x \frac{dv}{dx}$$

Put  $\frac{dy}{dx}$  in eq (i), we get,

$$v + x \frac{dv}{dx} = \frac{v^2 - 1}{2v}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{v^2 - 1}{2v} - v$$

$$\Rightarrow x \frac{dv}{dx} = \frac{v^2 - 1 - 2v^2}{2v}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{-1 - v^2}{2v}$$

$$\Rightarrow \int \frac{2v dv}{v^2 + 1} = \int \frac{-dx}{x}$$

$$\Rightarrow \log(v^2 + 1) = -\log x + c$$

$$\Rightarrow \log((v^2 + 1) \cdot x) = c$$

$$\Rightarrow (v^2 + 1) \cdot x = e^c$$

$$\Rightarrow \left(\frac{y^2}{x^2} + 1\right) \cdot x = e^c \quad \left[v = \frac{y}{x}\right]$$

$$\Rightarrow \frac{x^2 + y^2}{x} = A \quad [\because e^c = A]$$

$$\Rightarrow x^2 + y^2 = Ax$$

OR

$$\text{We have } \frac{dy}{dx} = \frac{x^2 + y^2}{x^2 + xy}$$

$$\text{Let } f(x, y) = \frac{x^2 + y^2}{x^2 + xy}$$

Here, putting  $x = kx$  and  $y = ky$

$$f(kx, ky) = \frac{(kx)^2 + (ky)^2}{(kx)^2 + kx \cdot ky} = k^0 \cdot f(x, y)$$

Therefore, the given differential equation is homogeneous.

$$(x^2 + xy)dy = (x^2 + y^2)dx$$

$$\frac{dy}{dx} = \frac{x^2 + y^2}{x^2 + xy}$$

To solve it we make the substitution.

$$y = vx$$

Differentiating above eq. with respect to  $x$ , we get

$$\frac{dy}{dx} = v + x \frac{dv}{dx}$$

$$v + x \frac{dv}{dx} = \frac{x^2 + (vx)^2}{x^2 + x \cdot vx}$$

$$v + x \frac{dv}{dx} = \frac{x^2(1 + v^2)}{x^2(1 + v)}$$

$$v + x \frac{dv}{dx} = \frac{1 + v^2}{1 + v}$$

$$x \frac{dv}{dx} = \frac{1 + v^2}{1 + v} - v = \frac{1 + v^2 - v - v^2}{1 + v}$$

$$x \frac{dv}{dx} = \frac{1-v}{1+v}$$

$$\frac{1+v}{1-v} dv = \frac{1}{x} dx$$

Integrating on both side,

$$\int \frac{1+v}{1-v} dv = \int \frac{1}{x} dx$$

$$\int \left( -1 + \frac{2}{1-v} \right) dv = \int \frac{1}{x} dx$$

$$-v - 2\log|1-v| = \log|x| + \log C$$

$$-\frac{y}{x} - 2\log\left|1 - \frac{y}{x}\right| = \log|x| + \log C$$

$$-\frac{y}{x} = 2\log\left|1 - \frac{y}{x}\right| + \log|x| + \log C$$

$$-\frac{y}{x} = \log \frac{(x-y)^2}{x^2} + \log|x| + \log C$$

$$-\frac{y}{x} = \log \frac{(x-y)^2}{x^2} \cdot Cx$$

$$-\frac{y}{x} = \log \frac{(x-y)^2}{x} C$$

$$-\frac{y}{x} = \log \frac{(x-y)^2}{x} C$$

$$\frac{C(x-y)^2}{x} = e^{-y/x}$$

$$C(x-y)^2 = xe^{-y/x}$$

$$(x-y)^2 = kxe^{-y/x}$$

Which is the required solution of the given differential equation.

13. Given:

$$\text{Curves } y^2 = 8x \dots (i)$$

$$\& 2x^2 + y^2 = 10 \dots (ii)$$

The point of intersection of two curves are (0, 0) & (1,  $2\sqrt{2}$ )

Now, Differentiating curves (i) & (ii) w.r.t. x, we get

$$\Rightarrow y^2 = 8x$$

$$\Rightarrow 2y \cdot \frac{dy}{dx} = 8$$

$$\Rightarrow \frac{dy}{dx} = \frac{8}{2y}$$

$$\Rightarrow \frac{dy}{dx} = \frac{4}{y} \dots (iii)$$

$$\Rightarrow 2x^2 + y^2 = 10$$

Differentiating above w.r.t. x,

$$\Rightarrow 4x + 2y \cdot \frac{dy}{dx} = 0$$

$$\Rightarrow 2x + y \cdot \frac{dy}{dx} = 0$$

$$\Rightarrow y \cdot \frac{dy}{dx} = -2x$$

$$\Rightarrow \frac{dy}{dx} = \frac{-2x}{y} \dots (iv)$$

Substituting (1,  $2\sqrt{2}$ ) for  $m_1$  &  $m_2$ , we get,

$$m_1 = \frac{4}{y}$$

$$\Rightarrow \frac{4}{2\sqrt{2}}$$

$$m_1 = \sqrt{2} \dots (v)$$

$$m_2 = \frac{-2x}{y}$$

$$\Rightarrow \frac{-2 \times 1}{2\sqrt{2}}$$

$$m_2 = -\frac{1}{\sqrt{2}} \dots (vi)$$

$$\text{When } m_1 = \sqrt{2} \& m_2 = \frac{-1}{\sqrt{2}}$$

Two curves intersect orthogonally if  $m_1 m_2 = -1$ , where  $m_1$  and  $m_2$  the slopes of the two curves.

$$\Rightarrow \sqrt{2} \times \frac{-1}{\sqrt{2}} = -1$$

$\therefore$  Two curves  $y^2 = 8x$  &  $2x^2 + y^2 = 10$  intersect orthogonally.

OR

Let ABC be right-circular cone having radius 'r' and height 'h'. If V and S are its volume and surface area (curved) respectively, then

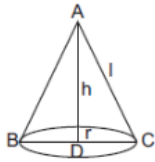
$$S = \pi r l$$

$$S = \pi r \sqrt{h^2 + r^2} \dots\dots (i)$$

Putting the value of h in (i), we get

$$S = \pi r \sqrt{\frac{9V^2}{\pi^2 r^4} + r^2}$$

$$\Rightarrow S^2 = \pi^2 r^2 \left( \frac{9V^2 + \pi^2 r^6}{\pi^2 r^4} \right)$$



$$\left[ \begin{array}{l} \because V = \frac{1}{3} \pi r^2 h \\ h = \frac{3V}{\pi r^2} \end{array} \right.$$

[Maxima or Minima is same for S or  $S^2$ ]

$$\Rightarrow S^2 = \frac{9V^2}{r^2} + \pi^2 r^4$$

$$\Rightarrow (S^2)' = \frac{-18V^2}{r^3} + 4\pi^2 r^3 \dots\dots (ii) \text{ [Differentiating w.r.t. 'r']}$$

Now,  $(S^2)' = 0$

$$\Rightarrow -18 \frac{V^2}{r^3} + 4\pi^2 r^3 = 0$$

$$\Rightarrow 4\pi^2 r^6 = 18V^2$$

$$\Rightarrow 4\pi^2 r^6 = 18 \times \frac{1}{9} \pi^2 r^4 h^2 \text{ [Putting value of V]}$$

$$\Rightarrow 2r^2 = h^2 \Rightarrow r = \frac{h}{\sqrt{2}}$$

Differentiating (ii) w.r.t. 'r', again

$$(S^2)'' = \frac{54V^2}{r^4} + 12\pi^2 r^2$$

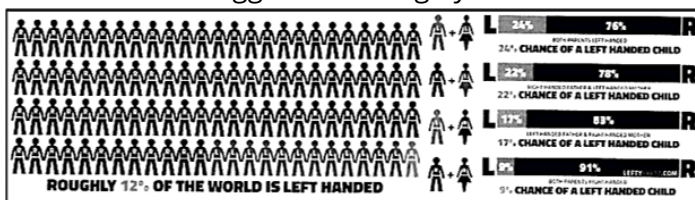
$$\Rightarrow (S^2)'' \Big|_{r=\frac{h}{\sqrt{2}}} > 0 \text{ (for any value of r)}$$

Hence,  $S^2$  i.e., is minimum for  $r = \frac{h}{\sqrt{2}}$  or  $h = \sqrt{2}r$ .

i.e., For least curved surface, altitude is equal to  $\sqrt{2}$  times the radius of the base.

#### 14. Read the text carefully and answer the questions:

Recent studies suggest that roughly 12% of the world population is left handed.



Depending upon the parents, the chances of having a left handed child are as follows:

- When both father and mother are left handed:  
Chances of left handed child is 24%.
- When father is right handed and mother is left handed:  
Chances of left handed child is 22%.
- When father is left handed and mother is right handed:  
Chances of left handed child is 17%.
- When both father and mother are right handed:  
Chances of left handed child is 9%.

Assuming that  $P(A) = P(B) = P(C) = P(D) = \frac{1}{4}$  and L denotes the event that child is left handed.

$$(a) P\left(\frac{L}{C}\right) = \frac{17}{100}$$

$$(b) P\left(\frac{L}{A}\right) = 1 - P\left(\frac{L}{A}\right) = 1 - \frac{24}{100} = \frac{76}{100} \text{ or } \frac{19}{25}$$

$$(c) P\left(\frac{A}{L}\right) = \frac{\frac{1}{4} \times \frac{24}{100}}{\frac{1}{4} \times \frac{24}{100} + \frac{1}{4} \times \frac{22}{100} + \frac{1}{4} \times \frac{17}{100} + \frac{1}{4} \times \frac{9}{100}} = \frac{24}{72} = \frac{1}{3}$$

(d) Probability that a randomly selected child is left-handed given that exactly one of the parents is left-handed.

$$= P\left(\frac{L}{B \cup C}\right) = \frac{22}{100} + \frac{17}{100} = \frac{39}{100}$$

### SECTION B - 15 MARKS

15. In subparts (i) and (ii) choose the correct options and in subparts (iii) to (v), answer the questions as instructed.

(a) **(d)**  $60^\circ$

**Explanation:**

$$\vec{a} \cdot \vec{b} = \frac{1}{2} |\vec{a}| |\vec{b}|$$

$$|\vec{a}| |\vec{b}| \cos \theta = \frac{1}{2} |\vec{a}| |\vec{b}|$$

$$\cos \theta = \frac{1}{2}$$

$$\theta = 60^\circ$$

(b) The direction ratios of the first line are (3, 5, 4) and the direction ratios of the second line are (1, 1, 2).

If  $\theta$  is the angle between them, then

$$\cos \theta = \left| \frac{3 \cdot 1 + 5 \cdot 1 + 4 \cdot 2}{\sqrt{3^2 + 5^2 + 4^2} \sqrt{1^2 + 1^2 + 2^2}} \right| = \frac{16}{\sqrt{50} \sqrt{6}} = \frac{16}{5\sqrt{2}\sqrt{6}} = \frac{8\sqrt{3}}{15}$$

Hence, the required angle is  $\cos^{-1}\left(\frac{8\sqrt{3}}{15}\right)$

(c) Let  $\vec{a} = 4\hat{i} - \hat{j} + 3\hat{k}$  and  $\vec{b} = -2\hat{i} + \hat{j} - 2\hat{k}$ . Then vector perpendicular to both vectors a & b is ;

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 4 & -1 & 3 \\ -2 & 1 & -2 \end{vmatrix} = (2 - 3)\hat{i} - (-8 + 6)\hat{j} + (4 - 2)\hat{k} = -\hat{i} + 2\hat{j} + 2\hat{k}$$

$$\Rightarrow |\vec{a} \times \vec{b}| = \sqrt{(-1)^2 + 2^2 + 2^2} = 3$$

$$\therefore \text{Required vector} = \frac{9\vec{a} \times \vec{b}}{|\vec{a} \times \vec{b}|} = \frac{9}{3}(-\hat{i} + 2\hat{j} + 2\hat{k}) = -3\hat{i} + 6\hat{j} + 6\hat{k}$$

(d) **(c)**  $90^\circ$

**Explanation:**

To find the angle with the z-axis, we use the relation:

$$\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$$

where  $\alpha = 30^\circ$  and  $\beta = 120^\circ$ . Calculating, we get:

$$\cos^2 30^\circ = \frac{3}{4}, \cos^2 120^\circ = \frac{1}{4}$$

Thus,

$$\frac{3}{4} + \frac{1}{4} + \cos^2 \gamma = 1 \Rightarrow \cos^2 \gamma = 0 \Rightarrow \gamma = 90^\circ.$$

So, the angle with the z-axis is  $90^\circ$ .

(e) d.r.'s of lines are  $\langle -2, 3p, 4 \rangle$  and  $\langle 4p, 2, -7 \rangle$

As lines are perpendicular

$$-8p + 6p - 28 = 0$$

$$\Rightarrow p = -14$$

16. We have,  $\vec{a} = \hat{i} + 2\hat{j} - 3\hat{k}$

$$\vec{b} = 3\hat{i} - \hat{j} + 2\hat{k},$$

$$\vec{a} + \vec{b} = 1 + 2\hat{j} - 3\hat{k} + 3\hat{i} - \hat{j} + 2\hat{k}$$

$$\Rightarrow \vec{a} + \vec{b} = 4\hat{i} + \hat{j} - \hat{k}$$

$$\vec{a} - \vec{b} = \hat{i} + 2\hat{j} - 3\hat{k} - (3\hat{i} - \hat{j} + 2\hat{k})$$

$$\Rightarrow \vec{a} - \vec{b} = -2\hat{i} + 3\hat{j} - 5\hat{k}$$

$$\text{Now } (\vec{a} + \vec{b}) \cdot (\vec{a} - \vec{b}) = (4\hat{i} + \hat{j} - \hat{k}) \cdot (-2\hat{i} + 3\hat{j} - 5\hat{k})$$

$$= (4 \times -2) + (1 \times 3) + (-1 \times -5) = -8 + 3 + 5 = 0$$

Here, we see that the dot product of two vectors is zero so the vector  $(\vec{a} + \vec{b})$  is perpendicular to  $(\vec{a} - \vec{b})$

Hence, proved.

OR

Let  $\theta$  be the angle between  $\vec{a}$  and  $\vec{b}$ , then

$$\cos \theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|} \dots (i)$$

$$\vec{a} \cdot \vec{b} = (\hat{i} - \hat{j})(\hat{j} + \hat{k})$$

$$= (\hat{i} - \hat{j} + 0 \times \hat{k})(0 \times \hat{i} + \hat{j} + \hat{k})$$

$$= (1)(0) + (-1)(1) + (0)(1)$$

$$= 0 - 1 + 0$$

$$\vec{a} \cdot \vec{b} = -1$$

$$|\vec{a}| = |\hat{i} - \hat{j}|$$

$$= |\hat{i} - \hat{j} + 0 \times \hat{k}|$$

$$= \sqrt{(1)^2 + (-1)^2 + (0)^2}$$

$$= \sqrt{1 + 1 + 0}$$

$$|\vec{a}| = \sqrt{2}$$

$$|\vec{b}| = |\hat{j} + \hat{k}|$$

$$= |0 \times \hat{i} + \hat{j} + \hat{k}|$$

$$= \sqrt{(0)^2 + (1)^2 + (1)^2}$$

$$= \sqrt{0 + 1 + 1}$$

$$|\vec{b}| = \sqrt{2}$$

put  $\vec{a} \cdot \vec{b}$ ,  $|\vec{a}|$  and  $|\vec{b}|$  in equation (i)

$$\cos \theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|}$$

$$= \frac{-1}{\sqrt{2} \times \sqrt{2}}$$

$$\cos \theta = \frac{-1}{2}$$

$$\theta = \cos^{-1} \left( -\frac{1}{2} \right)$$

$$\theta = \pi - \frac{\pi}{3}$$

$$\theta = \frac{2\pi}{3}$$

Therefore, the angle between  $\vec{a}$  and  $\vec{b} = \frac{2\pi}{3}$

17. Equation of one plane is  $\vec{r} \cdot (2\hat{i} + 2\hat{j} - 3\hat{k}) = 5 \dots (i)$

Comparing this equation with  $\vec{r} \cdot \vec{n}_1 = d_1$ , we have

$$\text{Normal vector to plane (i) is } \vec{n}_1 = 2\hat{i} + 2\hat{j} - 3\hat{k}$$

$$\text{Again, equation of second plane is } \vec{r} \cdot (3\hat{i} - 3\hat{j} + 5\hat{k}) = 3 \dots (ii)$$

Comparing this equation with  $\vec{r} \cdot \vec{n}_2 = d_2$ , we have

$$\text{Normal vector to plane (i) is } \vec{n}_2 = 3\hat{i} - 3\hat{j} + 5\hat{k}$$

Let  $\theta$  be the acute angle between planes (i) and (ii).

$\therefore$  angle between normals  $\vec{n}_1$  and  $\vec{n}_2$  to planes (i) and (ii) is also  $\theta$

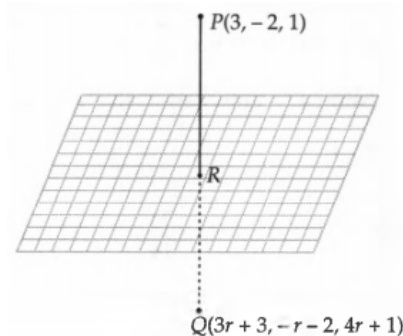
$$\begin{aligned}\therefore \cos \theta &= \frac{|\vec{n}_1 \cdot \vec{n}_2|}{|\vec{n}_1| |\vec{n}_2|} = \frac{|2(3) + 2(-3) + (-3)5|}{\sqrt{4+4+9} \sqrt{9+9+25}} \\ &= \frac{|6-6-15|}{\sqrt{17} \sqrt{31}} \\ &= \frac{|-15|}{\sqrt{731}} = \frac{15}{\sqrt{731}} \\ \Rightarrow \theta &= \cos^{-1} \frac{15}{\sqrt{731}}\end{aligned}$$

OR

Let Q be the image of the point P (3, -2, 1) in the plane  $3x - y + 4z = 2$ .

Then, PQ is normal to the plane. Therefore, direction ratios of PQ are proportional to 3, -1, 4. Since PQ passes through P (3, -2, 1) and has direction ratios proportional to 3, -1, 4.

Therefore, the equation of PQ is  $\frac{x-3}{3} = \frac{y+2}{-1} = \frac{z-1}{4} = r$



Let the coordinates of Q be (3r+3, -r-2, 4r+1). Let R be the mid-point of PQ.

Then, R lies on the plane  $3x - y + 4z = 2$ . The coordinates of R are

$$\left( \frac{3r+3+3}{2}, \frac{-r-2-2}{2}, \frac{4r+1+1}{2} \right) \text{ or, } \left( \frac{3r+6}{2}, \frac{-r-4}{2}, 2r+1 \right)$$

Since R lies on the plane  $3x - y + 4z = 2$

$$3 \left( \frac{3r+6}{2} \right) - \left( \frac{-r-4}{2} \right) + 4(2r+1) = 2 \Rightarrow 13r = -13 \Rightarrow r = -1$$

Putting  $r = -1$  in (3r + 3, -r - 2, 4r + 1), we obtain the coordinates of Q as (0, -1, -3)

Hence, the image of (3, -2, 1) in the plane  $3x - y + 4z = 2$  is (0, -1, -3).

18. To find area of region

$$\{(x, y) : 0 \leq y \leq x^2 + 3; 0 \leq y \leq 2x + 3; 0 \leq x \leq 3\}$$

$$\Rightarrow y = x^2 + 3 \dots (i)$$

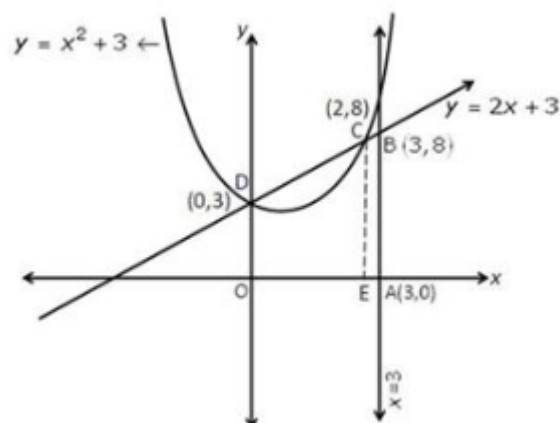
$$y = 2x + 3 \dots (ii)$$

and  $x = 0, x = 3$

Equation (i) represents a parabola with vertex (0, 3) and axis as y-axis.

Equation (ii) represents a line passing through (0, 3) and  $(-\frac{3}{2}, 0)$ .

A rough sketch of curve is as under:-



Thus Required area of the region = Area of the bounded Region ABCDOA

A = Region ABCEA + Region ECDOE

$$\begin{aligned}
&= \int_2^3 y_1 dx + \int_0^2 y_2 dx \\
&= \int_2^3 (2x + 3) dx + \int_0^2 (x^2 + 3) dx \\
&= (x^2 + 3x)_2^3 + \left(\frac{x^3}{3} + x\right)_0^2 \\
&= [(9 + 9) - (4 + 6)] + \left[\left(\frac{8}{3} + 2\right) - (0)\right] \\
&= [18 - 10] + \left[\frac{14}{3}\right] \\
&= 8 + \frac{14}{3} \\
A &= \frac{38}{3} \text{ sq. units.}
\end{aligned}$$

### SECTION C - 15 MARKS

19. In subparts (i) and (ii) choose the correct options and in subparts (iii) to (v), answer the questions as instructed.

(a) (c) ₹ 449

**Explanation:**

$$AC = \frac{Cx}{x} \Rightarrow AC = \frac{2000}{x} + 50 - \frac{1}{5}x$$

$$\text{Now, } [AC]_{x=5} = \frac{2000}{5} + 50 - \frac{5}{5} = ₹ 449$$

(b) (a) linear function

**Explanation:**

linear function

(c) Let the line of regression of y on x be

$$2x + 3y - 10 = 0$$

$$\Rightarrow 3y = -2x + 10$$

$$\Rightarrow y = -\frac{2}{3}x + \frac{10}{3}$$

$$\therefore b_{yx} = -\frac{2}{3}$$

Let the line of regression of x on y be

$$4x + y - 5 = 0$$

$$\Rightarrow 4x = -y + 5$$

$$\Rightarrow x = -\frac{1}{4}y + \frac{5}{4}$$

$$\therefore b_{xy} = -\frac{1}{4}$$

$$\text{Here, } b_{yx} \times b_{xy} = \left(-\frac{2}{3}\right) \left(-\frac{1}{4}\right) = \frac{1}{6} < 1$$

Which is true. Hence, our assumption is correct and line of regression of y on x is  $2x + 3y - 10 = 0$

(d) Let x be the number of items produced and sold. Let C(x) be the total cost incurred in producing x items. Then,

$$C(x) = \text{Fixed cost} + \text{Variable cost} = 26,000 + 30x$$

Let R(x) be the total revenue received in selling x items. Then,  $R(x) = 43x$

At the break-even point, we have  $C(x) = R(x)$

$$\Rightarrow 26,000 + 30x = 43x$$

$$\Rightarrow 13x = 26,000$$

$$\Rightarrow x = 2000$$

Hence, the break-even point is 2000 items.

(e) The demand function is linear, so assume  $p = a + bx$ . Using given points (1400, 4) and (1800, 2), solve for a and b.

$$\text{Slope } b = \frac{2-4}{1800-1400} = -\frac{2}{400} = -\frac{1}{200}$$

$$\text{Substituting } p = 4 \text{ at } x = 1400 : 4 = a - \frac{1400}{200}$$

$$a = 4 + 7 = 11 \text{ So, } p = 11 - \frac{x}{200}$$

$$\text{Revenue function: } R = p \cdot x = x \left(11 - \frac{x}{200}\right) = 11x - \frac{x^2}{200} \quad \text{Marginal revenue function:}$$

$$MR = \frac{dR}{dx} = 11 - \frac{x}{100}$$

Thus, the marginal revenue function is  $MR = 11 - \frac{x}{100}$ .

20. i. Since the demand function is given to be linear. So, let it be  
 $x = ap + b \dots (i)$   
 where  $p$  is the price per unit and  $x$  is the quantity demanded at this price.  
 When  $p = 4$ ,  $x = 1000$  and when  $p = 2$ ,  $x = 1500$   
 $\therefore 1000 = 4a + b$  and  $1500 = 2a + b$   
 Solving these two equations, we get:  $a = -250$  and  $b = 2000$   
 Putting the values of  $a$  and  $b$  in (i), we get  
 $x = -250p + 2000 \dots (ii)$   
 $\Rightarrow p = 8 - \frac{x}{250}$

Hence, the demand function is  $x = -250p + 2000$

ii. Let  $R$  be the total revenue function. Then,  $R = px \Rightarrow R = 8x - \frac{x^2}{250}$  [using (iii)]

iii. The average revenue function is  $AR = \frac{R}{x} = p \Rightarrow AR = 8 - \frac{x}{250}$  [using (iii)]

iv. The marginal revenue function is  $MR = \frac{dR}{dx} = 8 - \frac{x}{125}$  [using (iii)]

OR

Let  $C$  denote the cost and  $x$  the output. It is given that the cost  $C$  is a linear function of output  $x$ .

$\therefore C = ax + b$ , where  $a$ ,  $b$  are constants.

When  $x = 250$ , we have  $C = 4000$

$\therefore 4000 = 250a + b \dots (i)$

When  $x = 350$ , we have  $C = 5000$

$\therefore 5000 = 350a + b \dots (ii)$

Solving (i) and (ii), we get  $a = 10$  and  $b = 1500$ .

Substituting these values in  $C = ax + b$ , we get  $C = 10x + 1500$

21. Given,  $\bar{x} = 25$ ,  $\bar{y} = 30$ ,  $b_{yx} = 1.6$  and  $b_{xy} = 0.4$

i. Regression equation  $y$  on  $x$ :

$$y - \bar{y} = b_{yx}(x - \bar{x})$$

$$\Rightarrow y - 30 = 1.6(x - 25)$$

$$\Rightarrow y - 30 = 1.6x - 40$$

$$\Rightarrow y = 1.6x - 10 \dots (i)$$

ii. From eq. (i)

$$\text{When } x = 60, y = (1.6) \times (60) - 10$$

$$= 96 - 10 = 86$$

iii. Coefficient of correlation:

$$r = \pm \sqrt{b_{yx} b_{xy}}$$

$$= \pm \sqrt{(1.6)(0.4)}$$

$$= \pm \sqrt{0.64}$$

( $\because$  both regression coefficients are +ve)

22. i.

| Corner points | $Z = 3x - 4y$ |
|---------------|---------------|
| O(0, 0)       | 0             |
| A(0, 8)       | -32           |
| B(4, 10)      | -28           |
| C(6, 8)       | -14           |
| D(6, 5)       | -2            |
| E(4, 0)       | 12            |

Max  $Z = 12$  at E(4, 0)

Min  $Z = -32$  at A(0, 8)

ii. Since maximum value of  $Z$  occurs at B(4, 10) and C(6, 8)

$$\therefore 4p + 10p = 6p + 8p$$

$$\Rightarrow 2p = 2p$$

$$\Rightarrow p = q$$

Number of optimal solution are infinite.

OR

Let  $x$  kg of fertilizer type I and  $y$  kg of fertilizer type II be used, therefore the problem can be formulated as an L.P.P as under:

Minimize the cost (in Rs)  $Z = 0.6x + 0.4y$  subject to the constraints

$$\text{i.e. } \frac{10}{100}x + \frac{5}{100}y \geq 14 \text{ i.e. } 2x + y \geq 280$$

$$6\% \text{ of } x + 10\% \text{ of } y \geq 14$$

$$\text{i.e. } \frac{6}{100}x + \frac{10}{100}y \geq 14$$

$$\text{i.e. } 3x + 5y \geq 700$$

Draw the lines  $2x + y = 280$  and  $3x + 5y = 700$

and shaded the region satisfied by the above inequalities.

The feasible region convex and unbounded is shown in the figure. Therefore, we have,

The corner points are  $A\left(\frac{700}{3}, 0\right)$ ,  $B(100, 80)$  and  $C(0, 280)$

The values in  $z = 0.6x + 0.4y$  at the points A, B and C are 140, 92 and 112 respectively.

Among the values,  $z$  the least value is 92 as the feasible region is unbounded we draw the half has minimum value Rs 92 at the point B (100, 80). Therefore, the minimum cost = Rs 92, when 100 kg of fertilizer type 1 and 80 kg of fertilizer type II are mixed.

